



MILESTONE 1 · DUE 10 AUGUST 2026 · 20%

Project Charter Template

SAS821S CAPSTONE PROJECT · EXACTLY TWO GROUP MEMBERS

1. Group and project identification

Field	Group response
Topic ID and approved topic	T08 – Mining Remote Operations and Industrial-IoT Intrusion Analytics
Working project title	Security Analytics for Detecting Anomalous Behavior in Mining Remote Operations through a prototype
Member A	222093471 · Sophia Kalume · Data and Modelling Lead
Member B	215103815 · Lazarus Shithindi · Security Engineering and Intelligence Lead
Repository/location	TBC
Date submitted	10/08/2026

2. Problem and decision statement

The main problem and decision statement is “Identifying how security analytics can be used to detect and prioritise anomalous IIoT, network and remote-user behaviour in a simulated mining remote-operations environment”.

To monitor and manage operations, mining operations depend more on networked equipment, industrial Internet of Things (IIoT) devices, remote access and central control services. Although these technologies facilitate effective remote operations, the way they interconnected makes it possible for unauthorized access, anomalous network/device behavior and credential abuse. Unusual communication between devices, compromised remote-user credentials, anomalous sensor activity and other deviations from standard operating procedures can all be the source of suspicious behavior.



This project will focus primarily on IIoT telemetry, network-flow data and remote-user activity which will be analysed to establish normal behavioural patterns and identify significant deviations that may indicate potential security concerns. The main stakeholders affected are mining security personnel, IT ,network administrators, remote-operations personnel and operational managers as they require understandable analytical information that can assist them in determining which users devices or network events require further investigation.

The project also aims to identify which remote users, IIoT devices or network events should be prioritised for security investigation based on detected anomalous behaviour and associated risk and will be answered as the project progresses. A small segmentation simulation will additionally be used to demonstrate how network segmentation may affect the potential spread or impact of suspicious activity. This simulation will be performed in an authorised or simulated environment and will not involve testing live mining infrastructure.

3. Scope and exclusions

In scope	Out of scope
Simulated mining remote-operations environment, IIoT sensor telemetry, device telemetry, networkflow data, remote user activity, data cleaning, identification of abnormal behavioural patterns, IIoT and network anomaly detection, authorized, public or synthetic datasets.	Live mining infrastructure, physical industrial equipment, live network penetration testing, real user passwords or credentials, unauthorized or confidential data, real time employee surveillance, real-time production deployment.

4. Objectives and security analytics questions

Objective/question	Measure of completion
Objective 1: Prepare and integrate IIoT, network-flow and remote-user activity data for security analytics.	Cleaned datasets, documented preprocessing steps and identified analytical features.
Objective 2: Develop a prototype capable of detecting and prioritizing anomalous IIoT, network and remote-user behaviour.	Working anomaly-detection/behaviour-analysis process, risk prioritization and functioning prototype dashboard.



Objective/question	Measure of completion
Question 1: What IIoT and network behaviours deviate significantly from established normal patterns?	EDA and anomaly-detection results identifying and ranking anomalous events.
Question 2: Can remote-user activity be used to identify potentially unusual or suspicious access behaviour?	Behavioural indicators and identified anomalous user activities supported by evidence.
Question 3: How can detected anomalies be prioritized to support security investigation?	A documented risk-scoring approach and ranked security events displayed in the prototype.

5. Learning Sessions 1-10 coverage matrix

Session	Course area	Planned project evidence
1	Foundations of security analytics	Definition of the mining security problem, stakeholders, threats and security decision.
2	Analytical software and tools	Python-based data preparation, analysis and prototype development environment.
3	Machine-learning fundamentals	Feature preparation, model selection, training and evaluation principles.
4	Machine learning in cybersecurity	Application of anomaly detection to IIoT and network behaviour.
5	Analytics and incident response	Analysis and prioritisation of suspicious events for investigation.
6	Security intelligence	Conversion of analytical findings into risk indicators and prioritised security information.
7	Access analytics	Analysis of remote-user access patterns and unusual behaviour.
8	Simulations and security processes	Small segmented-network simulation comparing different security scenarios.
9	Security and text mining	Available maintenance/incident text data only if suitable authorised data is available.



Session	Course area	Planned project evidence
10	Advanced security intelligence	Integration of anomaly, user-behaviour and risk findings into the final prototype and security decision support.

6. Proposed data sources

Source	Format/volume	Key fields	Provenance/access	Security/ethics controls
IIoT telemetry	CSV/JSON, volume to be determined after dataset selection	Timestamp, device ID, sensor readings, operating status	Public, synthetic or approved dataset	No PII, only authorized data and anonymised device identifiers where necessary
Network-flow data	CSV/JSON/logs, volume to be determined after dataset selection	Timestamp, source, destination, protocol, port, packets, bytes, duration	Public, synthetic or approved dataset	No live network capture
Remote-user activity	CSV/JSON/logs, volume to be determined after dataset selection	User ID, timestamp, login/logout, source, device, access/action	Public, synthetic or approved dataset	No real credentials
Maintenance/incident text data	CSV/TXT, volume to be determined after dataset selection	Timestamp, equipment ID, event/maintenance description, category	Public, synthetic or approved data if suitable	No personal or confidential information

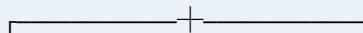


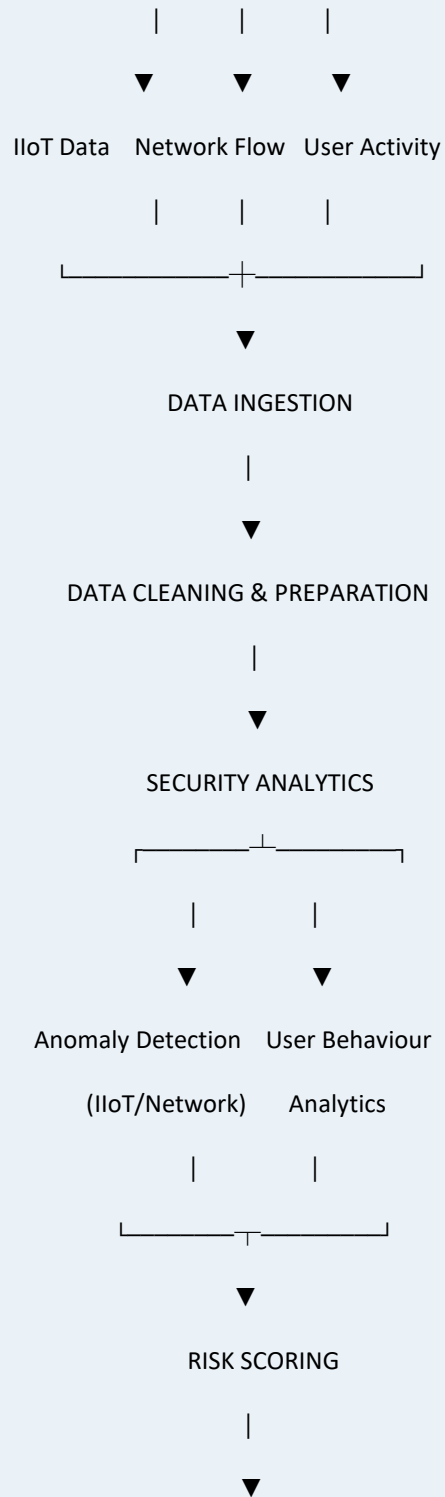
7. Proposed analytical methods

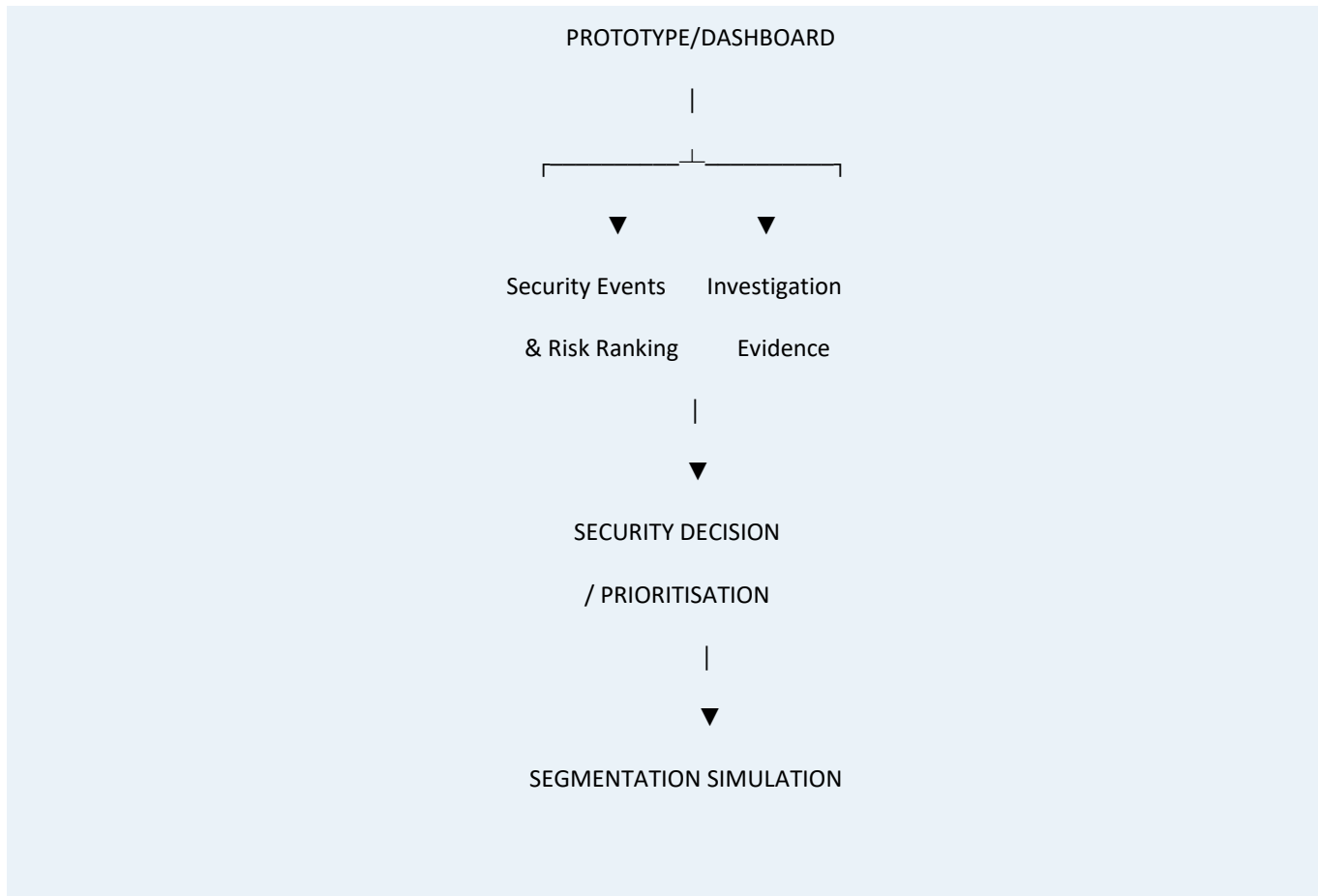
Component	Proposed method/tool	Expected output
Baseline / EDA	Python, Pandas, NumPy, visualization	Data-quality assessment, behavioural baselines, distributions and trends
Supervised ML	TBC	TBC
Unsupervised or UBA	Isolation Forest and rule/baseline-based user behaviour analysis	Anomaly scores and identification of unusual device/network/user behaviour
Incident investigation	Event filtering, correlation and timeline analysis	Evidence-based investigation view for selected suspicious events
Security intelligence	Risk scoring and event prioritization	Ranked high, medium and low-risk events
Simulation	Simple segmented-network scenario simulation	Comparison of potential event impact under different segmentation scenarios
Text mining/NLP	TBC	Supporting themes/indicators from maintenance or incident records
Predictive/adversarial analysis	Limited risk-pattern analysis	Identification of conditions associated with increased security risk
Prototype/dashboard	Python, Streamlit, Plotly	Interactive dashboard displaying anomalies, user behaviour, risk and investigation evidence

8. High-level architecture

DATA SOURCES







9. Roles and RACI

Workstream	Member A	Member B	Review/approval evidence
Data acquisition and quality	R/A	C	Dataset inventory, preprocessing documentation and data-quality results
ML and anomaly modelling	R/A	C	Model notebook, anomaly results and evaluation
Investigation and access analytics	C	R/A	User-behaviour analysis and investigation scenarios



Workstream	Member A	Member B	Review/approval evidence
Simulation and intelligence	C	R/A	Segmentation simulation and risk-prioritization results
Prototype and testing	R	A	Working dashboard and testing evidence
Documentation and presentation	R	R/A	Final report, presentation and contribution records

10. Work plan

Task	Owner	Start	Finish	Dependency	Output
Task 1: Finalize charter, research question and scope	Both	Week 1	Week 1	None	Approved project charter
Task 2: Identify, obtain and assess suitable datasets	Both	Week 1	Week 2	Task 1	Dataset inventory and suitability assessment
Task 3: Clean, integrate and explore datasets	Member A	Week 2	Week 3	Task 2	Prepared datasets and EDA results
Task 4: Develop anomaly and user-behaviour analytics	A/B	Week 4	Week 6	Task 3	Anomaly model and UBA results
Task 5: Develop risk scoring and investigation logic	Member B	Week 6	Week 7	Task 4	Risk-prioritization method
Task 6: Develop prototype dashboard and segmentation simulation	Both	Week 7	Week 8	Tasks 4–5	Working prototype and simulation



Task	Owner	Start	Finish	Dependency	Output
Task 7: Test, evaluate and refine prototype	Both	Week 8	Week 9	Task 6	Evaluation and revised prototype
Task 8: Final documentation and presentation preparation	Both	Week 9	Week 10	Task 7	Final report, presentation and project evidence

11. Risk and ethics register

Risk	Likelihood	Impact	Mitigation	Owner
Data unavailable or unsuitable	H	H	Assess datasets early, prioritise public/synthetic alternatives and adjust the analytical scope if necessary.	Both
Privacy or sensitive data exposure	L	H	Only authorised, public or synthetic data will be used anonymize identifiers and exclude credentials/PII.	Both
Model bias or misleading results	M	H	Establish behavioural baselines, validate results, compare outputs against known/synthetic anomalies and document limitations.	A/B
Scope/time overrun	H	H	Keep anomaly detection and user behaviour analytics as the core and make advanced methods conditional and review progress biweekly.	Both
Technical integration failure	M	M	Develop analytical components independently before integrating them into the dashboard and maintain version controlled backups.	A/B



Risk	Likelihood	Impact	Mitigation	Owner
Unequal contribution	M	M	Assign clear responsibilities, maintain contribution records and conduct regular progress reviews.	Both

12. Responsible AI and academic integrity statement

AI tools will be used as guidance and to get sources that may be needed for the project and this includes diagrams. The group will validate all outputs and disclose permitted assistance.

13. Group agreement and declaration

- We confirm that this group consists of exactly two students.
- We accept the role allocation and will retain evidence of our contributions.
- We understand that both members must explain the complete solution during the presentation and Q&A.
- We will use only authorised data and will not test live third-party systems.
- We will disclose permitted AI assistance, reused code and external data sources.

Member	Signature	Date
Sophia Kalume	S.Kalume	09/08/2026
Lazarus Shithindi	L.Shithindi	09/08/2026

14. Lecturer approval

Decision	Comments/conditions	Lecturer/date
[Approved / Revise / Not approved]	[Comments]	[Name/date]